

How Decisions Emerge: Action Dynamics in Intertemporal Decision Making

Maja Dshemuchadse, Stefan Scherbaum, and Thomas Goschke
Technische Universität Dresden

In intertemporal decision making, individuals prefer smaller rewards delivered sooner over larger rewards delivered later, often to an extent that seems irrational from an economical perspective. This behavior has been attributed to a lack of self-control and reflection, the nonlinearity of human time perception, and several other sources. Although an increasing number of models propose different mathematical descriptions of temporal discounting, the dynamics of the decision process behind temporal discounting are much less clear. In this study, we obtained further insights into the mechanisms of intertemporal decisions by observing choice action dynamics via a novel combination of continuously recorded mouse movements and a multiple regression approach. Participants had to choose between two hypothetical options (sooner/smaller vs. later/larger) by moving the mouse cursor from the bottom of the screen either to the top left or to the top right. We observed less direct mouse movements when participants chose later/larger rewards, indicating that participants had to overcome the attraction of the sooner/smaller reward first. Additionally, our results suggest that framing time information differently changes the weighting of value. We conclude that using a continuous process-oriented approach could further advance the understanding of intertemporal choice beyond the identification of the best fitted mathematical description of the discounting function by uncovering the way intertemporal decisions are performed.

Keywords: decision making, temporal discounting, intertemporal choice, action dynamics, date–delay effect

When facing significant decisions in their lives, people are commonly advised to resist first impulses and sleep on it. Studies of choice behavior support this skepticism about immediate impulses because these impulses often deviate from rationality as it is defined in rational choice theories adhering to normative economical rule such as utility maximization (Fishburn, 1968). A prominent example for the impact of impulsivity in decision theory is the preference for sooner over later delivered rewards, even if the sooner rewards are smaller in value. For these intertemporal decisions, the original discounted utility model assumes that the value of a delayed option decreases as an exponential function of the time until delivery (Samuelson, 1937). In contrast to this model, in empirical studies, researchers have found that individuals often discount rewards more steeply, especially for small time intervals or when the sooner reward is available immediately (e.g., Ainslie, 1975; Green, Fristoe, & Myerson, 1994; Laibson, 1997). The resulting discounting curves are better fitted by hyperbolic or quasihyperbolic functions (e.g., Berns, Laibson, & Loewenstein, 2007; Loewenstein & Prelec, 1992). These findings led to the

hypothesis that impulsivity forces people to discount future rewards more steeply than rational choice models would prescribe (Ebert & Prelec, 2007; Frederick, Loewenstein, & O'Donoghue, 2002).

In line with this hypothesis, clinical studies showed stronger discounting in patients with higher impulsivity scores, suffering from impulsivity related diseases such as addiction and attention-deficit/hyperactivity disorder (Bickel & Marsch, 2001; e.g., Wittmann & Paulus, 2008). Additionally, imaging studies support the role of impulsivity in intertemporal choice (McClure, Ericson, Laibson, Loewenstein, & Cohen, 2007; McClure, Laibson, Loewenstein, & Cohen, 2004; but see also Ballard & Knutson, 2009; Kable & Glimcher, 2007).

However, recent research raises several doubts as to whether impulsivity is the sole source of deviations from economic rationality. First, several findings indicate that time perception may be an important source of discounting (e.g., Loewenstein & Prelec, 1992; Wittmann, Leland, & Paulus, 2007). For example, assuming a logarithmic instead of a linear perception of temporal delays explains the observed hyperbolic discounting curves (Takahashi, 2005; Takahashi, Oono, & Radford, 2008; Zauberman, Kim, Malkoc, & Bettman, 2009). Second, studies showed reduced temporal discounting when the framing of the time information was changed from delays (e.g., “in 7 days”) to calendar dates (e.g., “on the 13th of November”), a phenomenon called the *date–delay effect* (LeBoeuf, 2006; Read, Frederick, Orsel, & Rahman, 2005). These considerations question the hypothesis that short-sighted choices are solely driven by spontaneous impulses. Instead, they suggest that temporal discounting may result primarily from how people perceive time intervals.

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Maja Dshemuchadse, Stefan Scherbaum, and Thomas Goschke, Department of Psychology, Technische Universität Dresden, Dresden, Germany.

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Correspondence concerning this article should be addressed to Maja Dshemuchadse, Department of Psychology, Technische Universität Dresden, Zellescher Weg 17, 01062 Dresden, Germany. E-mail: maja@psychologie.tu-dresden.de

Beyond the discussion of the specific contribution of impulsivity and time perception, several models present alternative explanations for temporal discounting, for example, utility discounting (Killeen, 2009), attribute-based weighing (Scholten & Read, 2010), temporal distance (Trope & Liberman, 2010), or time insensitivity (Ebert & Prelec, 2007). Because all of these models are constituted by mathematical functions derived from data based on the choice outcome (sooner/smaller vs. later/larger) under different conditions (varying value and time, varying frames, etc.) they are easy to compare (Doyle, 2010). Thus, it seems reasonable to evaluate them concerning goodness of fit and number of parameters to determine which model of delay discounting describes human choice behavior best. However, such an evaluation will only identify the best description of the observed choices, leaving open the question of how decisions are performed, that is, how humans or human brains implement this mathematical function.

In this article, we argue for the necessity to extend the focus from the final choice into the decision process itself to go beyond the search of the best fitting discounting function. By using a continuous approach that enables us to uncover the dynamics of the decision process (cf. Scherbaum, Dshemuchadse, & Kalis, 2008), we aim to improve the understanding of the mechanisms leading to temporal discounting and to shed light on the role of the two mentioned factors, impulsivity as a lack of reflection and time perception in temporal discounting.¹

A fruitful approach to studying the dynamics of decision processes has been used in recent studies (Freeman, Ambady, Rule, & Johnson, 2008; Scherbaum, Dshemuchadse, Fischer, & Goschke, 2010; Spivey, Grosjean, & Knoblich, 2005), in which participants made binary choices by moving a computer mouse to a right or left target location. The results showed that mouse movement trajectories were less direct when there was strong competition between alternative responses, providing a continuous measure of the influence of target and distracting information on the choice dynamics. Furthermore, this method also enabled the tracing of changes in mind during the decision process (Resulaj, Kiani, Wolpert, & Shadlen, 2009) and the identification of specific time windows in which different influences separately affected the decision process (Scherbaum et al., 2010).

In the present study, we applied this method to an intertemporal choice task for a first evaluation of a continuous method in approaching the mechanisms behind temporal discounting. Therefore, we address two assumptions underlying previous research.

First, we aimed to examine if choices for sooner/smaller rewards indeed involve less reflection than do choices for later/larger rewards. This should lead to different dynamics of the decision process and, more specifically to more direct mouse movements for sooner/smaller choices compared with the movements for later/larger choices. The indirectness of choice movements has been interpreted as a sign of prolonged reflection and changes of mind (Resulaj et al., 2009). To strengthen the hypothesis that more reflection is displayed in less direct choices, we compared choices with varying difficulty. Difficulty is maximal when the discounted value of the two choice alternatives is approximately equal (as indicated by the fact that participants choose both alternatives equally often, defining the *indifference point*). Choices closer to the indifference point are highly ambiguous and thus require more reflection, which should be indicated by less direct mouse movements, especially for later/larger choices.

Second, we wanted to explore the role of time perception during the decision process in intertemporal choices. Therefore, we examined the date–delay effect (Read et al., 2005; LeBoeuf, 2006) as an example for the influence of the presentation format of the time information on the amount of discounting. Although it has been demonstrated repeatedly that deciders exhibit less discounting when time is presented in calendar dates compared with the standard format of delays, the mechanisms behind this effect are still open. By observing the decision process continuously, we aimed to distinguish between three different explanations in two steps. In the first step, we distinguished two possible explanations: either reduced discounting in the date condition is just the general consequence of a more deliberative processing mode elicited by higher cognitive demands due to the more complex format of calendar dates; or it indeed reflects a difference in time perception. If the former was the case, we would expect increased reflection resulting in less impulsive and hence less direct mouse movements when time is framed in calendar dates in comparison to delays, especially for later/larger choices. Otherwise, one could assume that indeed a difference in time perception might be the reason for decreased discounting. In the second step, we distinguished between differences in the evaluation of time and the evaluation of value as two putative influences on the degree of temporal discounting. To obtain insight into how the decision was performed and to disentangle the two influences (time and value) during the decision process, we applied a novel regression analysis approach (Scherbaum et al., 2010). If the framing in calendar dates changed the way time was weighed, it should lead to a reduced influence of time in comparison to the delay condition. Alternatively, if the framing had a more indirect influence, it could instead change the way the value was weighed and, hence, lead to an increased influence of value in the date compared with the delay condition.

Taken together, the following study aims, on the one hand, to gain specific insight into the process of decision making leading to temporal discounting and, on the other hand, to demonstrate how an approach that extends the focus from the final choice to the decision process can enrich research on intertemporal choice.

Method

Participants

Forty-two right-handed students (28 women, mean age = 24.1 years) of the Technische Universität Dresden, Dresden, Germany, took part in the experiment. Participants gave informed consent to the study and received class credit or €5 (approximately \$6.50) payment. Six participants were excluded because they did not exhibit any discounting (k parameter of the fitted hyperbolic function < 0.01) in the different conditions (delay or date) and thus did not execute a sufficient number of sooner/smaller choices. An additional analysis including all participants did not qualitatively change the pattern of results.

¹ In this article, we use the term *lack of reflection* in a wide sense to denote an insufficient processing of choice-relevant information. This leaves open the issue of the extent to which these processes must be accessible to consciousness.

Apparatus and Stimuli

Stimuli were presented in white on a black background on a 17-in. screen running at a resolution of $1,280 \times 1,024$ pixels (75-Hz refresh frequency). The decision options were presented in the vertical center of the left and the right half of the screen. As targets for mouse movements, response boxes were presented at the top left and top right of the screen.

We used Psychophysics Toolbox 3 (Brainard, 1997; Pelli, 1997) in Matlab 2006b (MathWorks Inc., Natick, MA) as presentation software, running on a Windows XP SP2 personal computer. Participants performed their responses with a standard computer mouse (Logitech Wheel Mouse USB). Mouse movement trajectories were sampled with a frequency of 92 Hz and recorded from presentation of the complete choice options (including times and values) until the cursor reached the response boxes and the trial ended.

Procedure

Participants were asked to decide on each trial which of two options they preferred: the left (sooner/smaller) or the right (later/larger) option. Participants were instructed to respond to the hypothetical choices as if they were real choices. For reasons of practicability, we used hypothetical choices, as previous studies showed that they are comparable to real choices (e.g., Lagorio & Madden, 2005; Madden, Begotka, Raiff, & Kastern, 2003; Madden et al., 2004).

Trials were grouped into miniblocks of 14 trials (see Figure 1). Within each miniblock, the two option values remained constant and only the delay of the two options was varied. This approach was chosen to enable participants to respond fast enough with one continuous mouse movement. At the start of each miniblock, the values of the two options were presented for 5 s, allowing participants to encode the values in advance (see also Figure 1).

Each trial consisted of three stages. In the first stage, participants had to click a red box at the bottom of the screen (within a time limit of 1.5 s). This served to guarantee a comparable starting area for each trial. After participants clicked on this box, the second stage started. Two response boxes at the right and left upper corner of the screen were presented. Participants were instructed to start the mouse movement upward within a time limit of 1 s after the presentation of the response boxes. Only after moving at least four pixels in each of two consecutive time steps within the 1-s time limit, the third stage of the trial started. The two choice options appeared with the delay of each option presented beneath the value, either as a delay in days (e.g., “3 Tage”) or as a German calendar date (e.g., “19.04.2009”). The trial ended after the participant moved the cursor into one of the response boxes or after a time limit of 2 s. We chose the procedure of forcing participants to be already moving when entering the decision process to assure that they did not decide first and then only executed the final movement (Scherbaum et al., 2010). If participants missed any time limit of one of the three stages, the next trial started with the presentation of the start box (see Figure 1).

We orthogonally varied the interval between the options (1, 2, 3, 5, 7, 10, and 14 days), the value of the sooner option as a percentage of the value of the later option (20%, 50%, 70%, 80%, 88%, 93%, 97%, and 99%), and the framing of the time (delay vs. date). The framing of the time was varied between blocks, balanced between participants; the percentage of the value of the option was varied between miniblocks; and the interval was varied between the options across trials. Additionally, we orthogonally varied the delay of the sooner option (0 and 7 days) and the value of the late option (€19.68 and €20.32, approximately \$26 and \$27, respectively). The delay of the sooner option and the value of the late option were varied to control for specific effects and to collect more data without repeating identical trials, which could have led

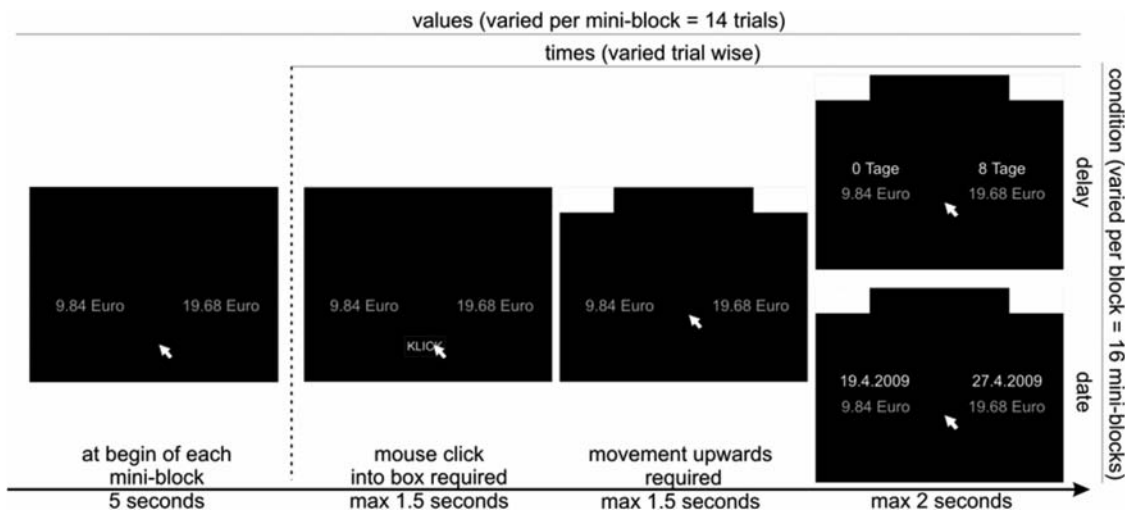


Figure 1. Procedure and setup of the experiment. The experiment was split up into blocks (varying the condition date/delay), consisting each of 16 miniblocks (varying option values), consisting in turn of 14 trials (varying option delays). Before each miniblock, the option values were presented for 5 s. Within each miniblock, participants had to click into a box at the bottom of the screen to start each trial. After this, they had to move the mouse cursor upwards. After reaching a movement criterion, the option delays were shown and participants had to choose an option by moving the mouse cursor into the top left or top right response box.

to memory effects. These factors had no significant effects and were excluded from following analyses. Altogether, the experiment consisted of two blocks (date and delay), order balanced across participants, and 224 trials (16 miniblocks) per block in randomized order.

Data Processing

To examine the date–delay effect, we determined individual discounting functions in two steps. First, we identified the indifference point, that is, the value difference for a particular time interval where a given subject chose indifferently between the two options. As an estimate of the indifference point, we determined the point of inflection of a logistic function fitted to the individual choices (sooner/smaller vs. later/larger) as a function of increasing value differences.² In the second step, we fitted for each subject a hyperbolic function to the indifference points over the different intervals and extracted the k parameter of this function (Green et al., 1994).³ We chose the hyperbolic function and the k parameter because the hyperbolic model has been widely used in other studies of discounting (e.g., Ballard & Knutson, 2009; Kable & Glimcher, 2007) and offers a parsimonious summary of discounting in a single parameter. Hence, it serves our aim to characterize decision behavior quantitatively rather than comparing different models (for a range of alternative models, see Doyle, 2010).

Trials on which participants did not reach a response box within the 2-s time limit (2%) and trials with movement onset times below 0.3 s (2%) were excluded as errors. Mouse trajectories were aligned for common starting position (horizontal middle position), and each trajectory was time normalized to 100 equal time slices to make the trajectories of different length comparable and analyzable (Spivey et al., 2005).

Results

Choice Data

As expected, participants showed more temporal discounting in the delay compared with the date condition (see Figure 2). A repeated-measures analysis of variance (ANOVA) with the independent variables time interval and condition (delay vs. date) and the dependent variable indifference point revealed significant main effects for time interval, $F(1, 35) = 78.2, p < .001$, and condition, $F(1, 35) = 11.9, p < .01$ as well as a significant interaction, $F(6, 210) = 4.1, p < .001$. Likewise, the k parameter of the hyperbolic discounting curve was significantly higher in the delay condition ($k = 0.15$) than in the date condition ($k = 0.10$), $t(35) = 2.2, p < .05$, indicating stronger discounting in the delay condition.

Mouse Trajectories

Following previous studies on mouse trajectories (e.g., Spivey et al., 2005), we calculated for each trial the degree of curvature of the movement trajectory. Curvature is defined as the area between each trajectory and a straight line from the start point to the end point of this trajectory. Hence, more direct mouse movements are reflected in a smaller curvature and more indirect movements are reflected in an increased curvature.⁴

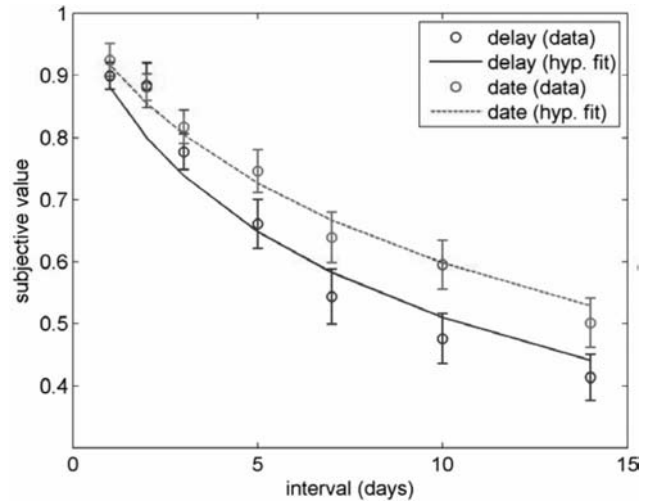


Figure 2. Indifference points depicting the decrease in subjective value as a function of intervals between the two options for the two conditions delay and date. The curves display the fitted hyperbolic (hyp.) functions for the conditions delay and date. Error bars indicate standard errors.

An ANOVA with the independent variables condition (delay vs. date) and choice (sooner/smaller vs. later/larger) revealed a significant main effect of choice, $F(1, 35) = 4.5, p < .05$; an insignificant main effect of condition, $F(1, 35) = 0.2, p = .7$; and an interaction of condition and choice, $F(1, 35) = 5.2, p < .05$ (see Figure 3).

Trajectories on trials in which participants chose the later but larger option showed more pronounced curvature than did trials on which participants chose the sooner option, that is, the mouse movements to the response box were less straight (see Figure 4). This difference was larger in the delay condition and smaller in the date condition.

Additionally, we analyzed the influence of the decision difficulty on the degree of curvature. We operationalized decision difficulty as the distance of each decision from the respective indifference point (point of indifference between the two options). Decisions far from the indifference point (i.e., if, for a given time interval, the value difference between the two choice options is much higher or lower than the value difference at the indifference point) should be easier than decisions close to the indifference

² The fitting of the logistic regression model was performed using the StixBox mathematical toolbox by Anders Holtsberg (<http://www.maths.lth.se/matstat/stixbox/>). The fit was based on the model $\log[p/(1 - p)] = X \times b$, where p is the probability that the choice is 1 (sooner option) and not 0 (later option), X represents value differences, and b represents the point estimates for the logistic function.

³ The fitting of the hyperbolic function was performed by applying Matlab's multidimensional unconstrained nonlinear minimization function to the hyperbolic function $1/(1 + k \times x) = y$, with x denoting time interval, y denoting subjective value, and k denoting the discounting parameter.

⁴ To control for multimodality, we calculated bimodality coefficients (cf. Spivey et al., 2005) for each choice. For the z distribution over sooner/smaller choices, bimodality coefficient (b) was 0.14, and for the z distribution over later/larger choices, it was 0.13 (with $b > 0.555$ being the standard cutoff for multimodality).

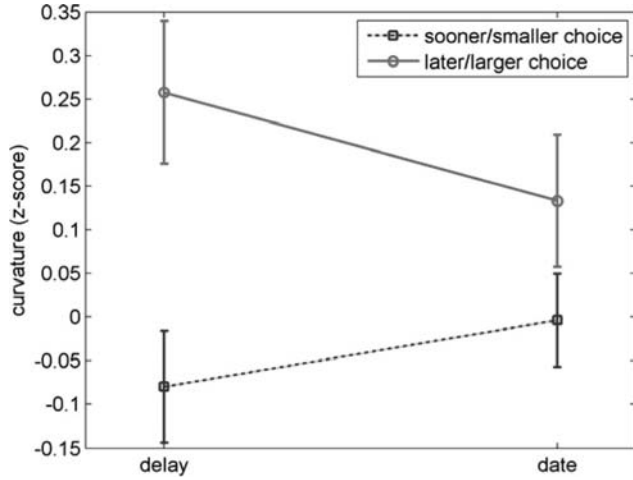


Figure 3. Mean area under the curve of the trajectories for the two choices sooner/smaller versus later/larger under the two conditions delay and date. Larger values indicate less direct choices. Error bars indicate standard errors.

point. We therefore divided the trials depending on their distance to the subject's individual indifference point into quartiles, and then we compared the curvature across quartiles. An ANOVA on curvature with the independent variables difficulty and choice (sooner/smaller vs. later/larger) revealed significant main effects of difficulty, $F(3, 35) = 40.8, p < .001$, and choice, $F(1, 35) = 5.9, p < .05$, and a significant interaction, $F(3, 105) = 28.4, p < .001$ (see Figure 5). Decisions that were close to the indifference point and hence more difficult showed more curvature than did less ambiguous decisions far from the indifference point, especially when the later/larger reward was chosen.

For further analyses, we focused on the trajectory angle on the XY plane. Trajectory angle was calculated as the angle relative to

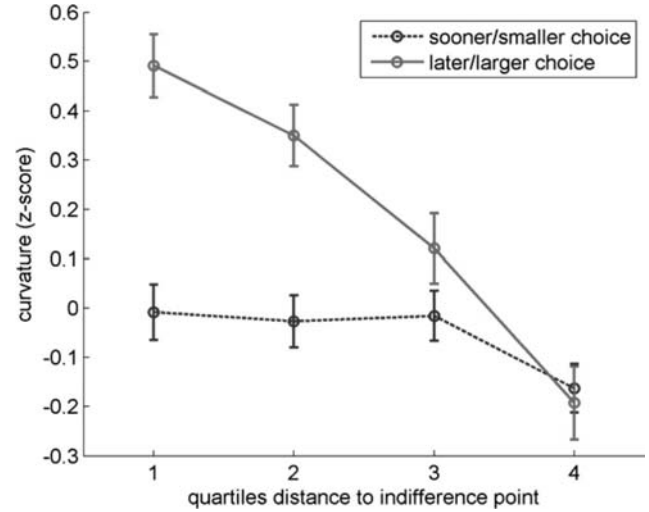


Figure 5. Mean area under the curve of the trajectories as a function of the distance from the indifference point (quartiles). The indifference point defines the subjective value where sooner/smaller and later/larger options are equal. Hence, quartiles with lower distances from the indifference point represent more difficult decisions. Error bars indicate standard errors.

the y-axis for each difference vector between two time steps. This measure has two advantages over the raw trajectory data. First, it better reflects the instantaneous tendency of the mouse movement because it is based on a differential measure rather than the cumulative effects in raw trajectory data. Second, it integrates the movement on the XY plane into a single measure. Given that it was our aim to dissect the influences of the independent variables on mouse movements within a trial, we applied a three-step procedure to the trajectory angle separately for the date and the delay condition (cf. also Scherbaum et al., 2010). In the first step, we coded for each participant two predictors for all trials: value difference

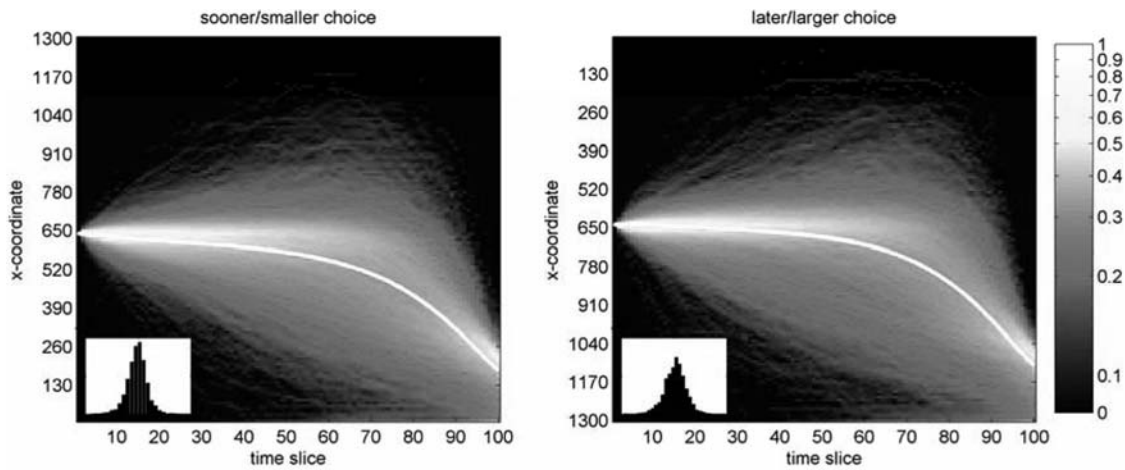


Figure 4. Probability (log transformed) plots of mouse trajectories on the x-axis pooled over all trials and participants for the two choices sooner/smaller versus later/larger. The white curve in each plot is the mean trajectory over all trials and participants of the choice. Insets show the distribution of the area under the curve over participants with the z score of the area under the curve on the x-axis $[-5, 5]$ and the relative frequency over participants on the y-axis $[0, 0.2]$.

(eight steps) and time interval (seven steps). To derive comparable beta weights in the next step, we normalized the values of the predictors to a range of -1 to 1 (e.g., the value difference predictor had eight different values ranging from -1 to 1). In the second step, we computed multiple regressions with these predictors (100 time slices, hence 100 multiple regressions) on the trajectory angle, which had also been standardized for each participant from -1 to 1 . This yielded two time-varying beta weights ($2 \text{ weights} \times 100 \text{ time slices}$) for each participant. Finally, in the third step, we computed grand averages of these two time-varying beta weights, yielding a time-varying strength of influence curve for each predictor in the two conditions (see Figures 6A and 6B).

To examine differences between the date and delay conditions, we compared the beta weights from the two conditions for each predictor and each time slice with paired t tests (see Figure 6C). To compensate for multiple comparisons of temporally dependent data, we chose as a criterion of reliability a minimum of eight consecutive significant t tests (see Dale, Kehoe, & Spivey, 2007, for Monte Carlo analyses on this issue).

The beta weights for the predictor *value difference* were reliably higher in the date condition compared with the delay condition in Time Slices 44–74 ($M_{\text{time}} = 386 \text{ ms} - 648 \text{ ms}$, all $t(20) > 2.0$, $p < .05$). The greater impact of the values in the date condition compared with the values in the delay condition matches well the stronger discounting found for delays compared with dates.

Discussion

In this study, we assessed the differential contributions of the two factors lack of reflection and time perception to temporal discounting. To open a window to the processes underlying intertemporal decisions, we recorded continuous mouse movements by which participants indicated their decisions in an intertemporal decision task. Detailed analyses of the resulting mouse trajectories revealed the influences of a lack of reflection and of the different

option properties (value and time) on participants' choices depending on the time framing of the options in terms of dates or delays.

First, we obtained evidence for the influence of a lack of reflection in intertemporal choice. The curvature of mouse trajectories revealed significant differences between sooner/smaller and later/larger choices: When participants chose later/larger rewards, the mouse was moved to the response box less directly compared with when participants chose sooner/smaller rewards. This effect was even stronger for the delay condition compared with the date condition. In line with our assumption that increased reflection shows up in less direct choices, we also found a higher curvature of mouse trajectories for more difficult decisions, especially when they resulted in the choice of later/larger rewards. Together, this indicates that later/larger choices required a longer reflection process, probably to overcome the attraction of the sooner reward. It should be noted that this finding does not imply the existence of two distinct systems, an "impulsive" and a "rational" one as proposed previously (e.g., McClure et al., 2004, 2007). Instead, the difference between sooner/smaller and later/larger decisions with regard to the degree of reflection and the amount of redecisions could also be attributed to functional differences in the information accumulation process within one system, leading to different initial choices on the way to the final choice (cf. Resulaj et al., 2009).

Second, our results suggest that a change in the weighting of the value information is a possible source of the date–delay effect (Read et al., 2005). Replicating previous findings, we found stronger temporal discounting when the time intervals of the options were presented as delays rather than calendar dates. By observing the decision process continuously, we distinguished between three different explanations in two steps. In the first step, we found that the difference between the curvature of the mouse movement trajectories of sooner/smaller and later/larger choices was smaller in the date condition than in the delay condition, combined with no general difference between choices in the delay and date condi-

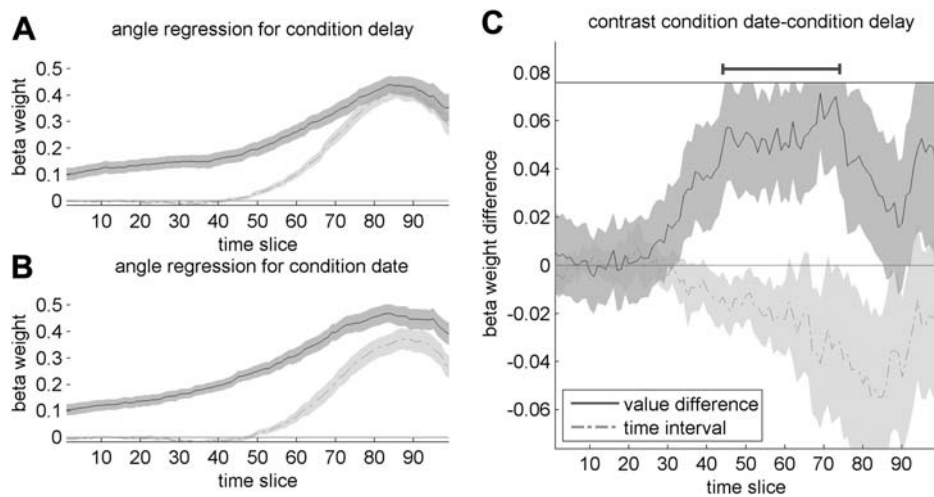


Figure 6. Beta weights and contrasted beta weights from regression analyses over time, dissecting the mouse trajectory angle on the XY plane (shaded areas around the curves indicate the standard error of beta weights for each time slice). A: Beta weights for each regressor over time for the condition delay. B: Beta weights for each regressor over time for the condition date. C: The contrast date–delay of the beta weights shown in A and B. Above the graph, time slices with a significant contrast between the two conditions are marked for each regressor.

tions. This speaks against the hypothesis that the date–delay effect is just the general consequence of a more deliberative processing mode elicited by higher cognitive demands due to the more complex format of calendar dates. If this was the case, we would have expected less direct mouse movements when time is framed in calendar dates in comparison to delays, especially for later/larger choices. In the second step, we found that the decision process was more strongly influenced by value differences in the date condition in comparison to the delay condition. Thus, our findings stand in contrast to the assumption that framing the time information differently changes the way time is weighed. Instead, they can be taken as evidence for framing changing the decision process in a more indirect way, leading to an increased influence of value in the date compared with the delay condition. As the difference between sooner/smaller and later/larger choices in the degree of curvature is reduced, we assume that choices are more similar with regard to the amount of reflection in the date condition. However, this finding does not provide direct evidence against the role of time perception in intertemporal choice models in general (e.g., Zauberman et al., 2009), because nonlinear time perception could still contribute to temporal discounting. In any case, the more indirect influence of the framing manipulation might indicate an even more complex decision process being at work despite a relatively simple underlying structure of subjective value (cf. Busemeyer & Johnson, 2004).

The presented results offer specific insight into the process of decision making leading to temporal discounting. Although we hope that this encourages further research, we nevertheless have to address several doubts concerning the validity and the value of our continuous, process-oriented approach.

First, we interpreted more indirect mouse movements—for example, for later/larger choices or more difficult choices—as a sign of a longer reflection process. However, especially in the latter case, less direct movements might also be interpreted as indecisiveness instead of enhanced reflection. Yet, the combined analysis of the influence of the final choice (soon or late option) and of trial difficulty (difference to the indifference point) on mouse trajectories revealed that especially choices of the later/larger option are affected by trial difficulty. If less direct movements were due to indecisiveness, both kinds of choices should have been affected by choice difficulty similarly. In contrast, our results suggest that enhanced reflection leads to choices of the later option, with more difficult decisions needing even more reflection to overcome the first impulse of choosing the sooner option. This interpretation matches previous studies indicating that trajectories constitute a useful online measure of the presence of conflicting response tendencies during the decision process (cf. Scherbaum et al., 2010; Spivey et al., 2005).

Second, our experimental design confounds the direction of mouse movements (left and right) with the position of the two options (sooner/smaller and later/larger). In line with previous temporal discounting studies (e.g., Figner et al., 2010; McClure et al., 2004; Read et al., 2005), we did not counterbalance the mapping of the options (left = soon, right = late) because of the typical number line arrangement (Dehaene, Bossini, & Giraux, 1993). However, this might be problematic because of differences, for example, in the motor kinematics when moving the mouse in the right hand to the left or to the right. Two observations provide evidence against this possible effect. First, the difference in curvature between sooner/smaller and later/larger choices almost van-

ished for very easy decisions, indicating no principle effect of motor kinematics due to the unbalanced setup. Second, we additionally analyzed data from a previous mouse movement study (Scherbaum et al., 2010) to exclude such motor kinematic effects in principle. In this study, participants had to perform a Simon task in which they had to respond to arrows pointing either to the left or to the right. In the data of this completely balanced setup, we did not find any differences between the mouse trajectories to the left and the right caused by the direction of response. However, researchers conducting future studies might be able to avoid the described difficulties, for example, by constructing intertemporal choice tasks where the options are shown on varying places (e.g., top/bottom) and not only left/right.

Third, although the presented results advance the understanding of intertemporal choice, the generalizability of our results provided by the continuous process-oriented approach certainly needs further convergent evidence from future studies, for example, by comparing our specific methodology with other methods used in the assessment of temporal discounting (cf. Frederick et al., 2002). Even though further research is needed for a final evaluation of our approach, our study adds to a growing body of research demonstrating the value of analyzing the dynamics of decision processes rather than focusing solely on behavioral outcomes of decision processes (e.g., Busemeyer & Townsend, 1993; Erlhagen & Schoner, 2002; Scherbaum et al., 2008).

Finally, the continuous process-oriented approach presented here could advance knowledge about intertemporal decision making in the following way. First, we expect that observing the continuous influence of different variables on the decision process will enable a systematic comparison between different models for temporal discounting. For example, it would be possible by systematic variations of time and value to observe their influence on the weighing process depending on their relative amount indicating either an alternative-based or an attribute-based weighing process (as proposed by Scholten & Read, 2010). Second, by extending the dynamical toolset, we could gain further insights into the decision process (Scherbaum et al., 2008). For example, one could investigate attentional processes concerning the different option properties over time (Busemeyer & Johnson, 2004; Townsend & Busemeyer, 1995), for example, by measuring the ongoing allocation of attention as reflected in frequency tagged electric brain potentials (Scherbaum, Fischer, Dshemuchadse, & Goschke, 2011). Thus, our continuous approach will go beyond the identification of the best fitting mathematical description of the discounting function by offering progress in the understanding of the way intertemporal decisions are performed.

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